A LIVE DEMONSTRATION

Agents as a review panel

Five specialist AI agents review a data management plan. We watch where they help, and where they go wrong.

Nick Owen · UCL Advanced Research Computing · [Event, date]

What is an agent?

A language model working in a loop towards a goal: it decides what to do, uses tools, looks at the result, and goes again until the job is done.



> Unlike a chatbot, an agent takes actions and sees their effects. That lets it do multi-step work, and it means its mistakes can compound.

A team is a set of job descriptions

Each agent is a short text file: its name, its remit, the tools it may use, and the form its answer must take.

Each works in its own context. It sees only what it is given, not what the others are thinking.

The files live with the project, under version control, so the team can review and improve them like any other document.

```
---
name: data-steward
description: Reviews a data management plan for
metadata, repositories, retention and costs.
tools: Read, Grep, Glob
---
You are the data steward on a panel reviewing a
data management plan…
```

Claude Code reads these files from **.claude/agents/** in the repository.

Why a team, and what can go wrong

What a team should add

Depth: each specialist knows one area well.

Independence: separate reviewers, separate blind spots.

Speed: reviews run in parallel.

What can go wrong

Conformity: drifting towards agreement.

Loss in summary: findings softened or dropped.

Giving way: backing down when pushed.

Invention: confident claims with no basis.

The Virtual Lab (Swanson et al., *Nature* 2025): an AI principal investigator led AI scientists through research meetings and designed 92 nanobodies, which were then tested in the lab. A limitation the lead author reported:

“They were too agreeable with each other.”

THE AIM

Does a team of specialists beat one generalist, and where does it go wrong?

Same plan, same reading pack, same checklist for both. Every run is scored against an answer key the agents never see.

What we measure

Measure	The question	When
Found	Did they spot the known problems?	Every stage
Invented	Claims with no basis in the plan or reading pack	Every stage
False alarms	Flagging things that are actually fine	Stages 1 and 2
Conformity	Changing a finding without new evidence	Stage 3
Lost in write-up	Findings dropped or softened by the report	Stages 4 and 5
Holding ground	Keeping a sound finding under pushback, and dropping a weak one	Stage 7

The case: MN-TRANSCRIPT

A data management plan for an MRC-funded study at the fictional University of Northbridge.

- Motor neurons grown from stem cells of people with C9orf72 ALS and of healthy volunteers
- Whole-genome sequencing, RNA-seq and NHS clinical data
- A US collaborator, public data deposit and a trusted research environment

It contains known problems, and a few traps that look wrong but are not.

The chair holds the answer key. The agents never see it.

The laws and guidance are real. The people, institution and study are invented.

The panel

Specialists: review the plan

Data steward: metadata, repositories, retention, costs

IG officer: UK GDPR, DPIA, sharing, transfers

Bioinformatician: design, data volumes, pipelines

Ethics reviewer: approvals, consent, genetic findings

TRE officer: the secure environment, data in and out

Process roles: run and check

Generalist: the baseline, one agent doing everything

Secretary: writes the report from the findings alone

Critic: checks the panel's evidence and the report

Applicant: pushes back on chosen findings

Chair: runs the stages and saves every output, but writes no findings

How the panel fits together

Chair The main Claude Code session. Runs each stage from CHAIR.md, passes files between agents, saves every output, writes no findings.

INPUTS

The plan
dmp.md and samplesheet.csv

Reading pack
The only sources agents may cite

REVIEWERS

Generalist
Baseline, reviews alone

Five specialists

- Data steward
- IG officer
- Bioinformatician
- Ethics reviewer
- TRE officer

REGISTER

Findings register

`round1.json`
`round2.json`
`rebuttal.json`

Specialists read it back in the discussion and rebuttal rounds

OUTPUT AND CHECKS

Secretary
Writes the report

`Trace check`
A script, not an AI

Critic
Checks the panel

Applicant
Pushes back

Ground rules for every agent

Read-only

They can read files. They cannot edit, run code or browse the web.

Reading pack only

They may cite only the plan and the pack: digests of real UK guidance, plus two fictional University policies.

Evidence required

Each finding quotes its source, 30 words at most, or shows its working.

One register

Findings are saved as structured data, so every step can be checked.

Reasons to change

A finding changes only with new evidence, not because others disagree.

Safe by design

A fictional study, no real data, run in an isolated cloud sandbox.

Anatomy of a finding

Field	What it holds
ID	Role and number, such as STW-1
Plan section	Where the problem is
Issue	What is wrong, and why it matters
Evidence	A quote from the reading pack, or the working
Severity	Major or minor
Recommendation	What would fix it
Confidence	High, medium or low

The ID is the thread through the whole session.

Every sentence in the final report must cite the IDs it comes from, so any claim can be traced back to its evidence.

Eight stages

1

Baseline

One generalist reviews everything

2

Independent review

Five specialists, in parallel

3

Discussion

They see each other’s findings

4

Write-up

A secretary drafts the report

5

Trace check

A script, not an AI

6

Critic

Checks the evidence and the report

7

Rebuttal

The applicant pushes back

8

Close

Score against the answer key

STAGE 1 OF 8 · ABOUT 5 MINUTES

Baseline: one generalist

What happens

One generalist agent reviews the whole plan alone, using all five specialists' checklists combined.

Same plan, same reading pack, same evidence rule as the panel.

Its findings are kept out of the register, so the specialists never see them.

Watch for

Breadth against depth: does it cover every area, and how deep does it go in each?

We record

Known problems found, invented findings, false alarms.

STAGE 2 OF 8 · ABOUT 10 MINUTES

Independent review: five specialists

What happens

The five specialists review the plan in parallel. Each works in its own context and cannot see the others.

Anything outside a specialist's remit is passed on as a referral, not raised as a finding.

The chair saves all five reviews, unchanged, to the register.

Watch for

Overlaps, referrals between specialists, and whether each quote really supports its point.

We record

Known problems found by each specialist, invented findings, false alarms.

STAGE 3 OF 8 · ABOUT 10 MINUTES

Discussion: seeing each other’s work

What happens

Each specialist reads everyone’s first-round findings, plus any referrals addressed to it.

It re-issues its findings, marking each as kept, revised, withdrawn or new.

A change needs new evidence. Other reviewers’ views do not count.

Watch for

Conformity: findings withdrawn or softened only because nobody else raised them.

We record

Changes made without new evidence.

STAGE 4 OF 8 · ABOUT 5 MINUTES

Write-up: the secretary's report

What happens

A secretary agent writes the report from the findings register alone. It never sees the plan.

Every sentence must cite the finding IDs it comes from, and nothing new may be added.

Disagreements are reported as disagreements, not resolved.

Watch for

Softened severity, findings merged that were different, and findings that quietly disappear.

We record

Which known problems survive into the report.

STAGE 5 OF 8 · ABOUT A MINUTE

Trace check: a script, not an AI

What happens

A short script compares the report with the register in both directions.

It lists IDs the report cites that do not exist, findings the report leaves out, withdrawn findings still cited, and sentences with no citation.

Watch for

Anything the secretary invented or dropped. A script gives the same answer every time.

We record

Invented and dropped IDs.

STAGE 6 OF 8 · ABOUT 5 MINUTES

Critic: checking the panel

What happens

The critic reviews the panel's work, not the plan.

Does each quote exist where it says, and support the point? Does the reasoning hold? Did the report distort any finding?

It flags conformity in the discussion round and names the weakest finding.

Watch for

Whether it catches what we noticed ourselves, and what it misses.

We record

Evidence problems, report distortions and conformity flags.

STAGE 7 OF 8 · ABOUT 10 MINUTES

Rebuttal: the applicant pushes back

What happens

The applicant agent challenges findings the chair chooses: one sound, one weak.

Each challenged specialist upholds, revises or withdraws, on the evidence alone.

Seniority, confidence and the cost of a fix are not evidence.

Watch for

Giving way: a sound finding dropped because the pushback sounded confident.

We record

Whether the sound finding is upheld and the weak one withdrawn.

STAGE 8 OF 8 · ABOUT 10 MINUTES

Close: scoring the run

What happens

The chair writes a summary of every stage’s counts from the saved files.

We score the run against the answer key and reveal the known problems.

Then we compare the generalist with the panel, and each stage with the one before.

Watch for

Where the team added value, and where a person had to step in.

We record

The scoreboard on the next slide.

Scoreboard

Measure	Baseline	Panel round 1	After discussion	In report	After rebuttal
Known problems found (of 12)					n/a
Invented findings					n/a
False alarms			n/a	n/a	n/a
Changed without new evidence	n/a	n/a		n/a	n/a
Lost in write-up	n/a	n/a	n/a		n/a
Held under pushback	n/a	n/a	n/a	n/a	

Questions to ask of any AI-assisted review

Where is the evidence?

Can each claim be traced to a source or a piece of working?

Who checked the summary?

A report can drop or soften what the reviewers found.

Why did it change its mind?

Agreement and confident pushback are not evidence.

What could a script check?

Use deterministic checks wherever they will do the job.

The expert stays in the loop: agents widen the search, and people judge the result.

OVER TO YOU

Discussion

Where would you trust a panel like this, and what would you need to see first?

Swanson K, Wu W, Bulaong NL, Pak JE, Zou J (2025). The Virtual Lab of AI agents designs new SARS-CoV-2 nanobodies. *Nature*. doi:10.1038/s41586-025-09442-9

UK Research Integrity Office (2025). Embracing AI with integrity: a practical guide for researchers.